

Final Project

Team registration due date: 02/10 (Tue)
Abstract (5%) due date: 03/03 (Tue)
Midpoint report (5%) due date: 04/09 (Thu)
Presentation (10%) due date: 04/28 and 04/30
Final report (20%) due date: 05/08 (Thu)

Project

- Should be done in teams of 3
- Exceptions (solo, duo, 4, 5) are permitted upon request with the [form](#).

There're four-steps:

1. Team registration, or a request for a team match (together with team registration/request for a team match for A3) – Due 02/10 (Tue)
 - a. We announce team matching by 02/17 (Tue)
2. (Checkpoint 1) Abstract submission (<250 words): Students should think about the topic in advance, and check if the topic is out of the scope – Due 03/03 (Tue) We'll also use this information for assigning breakout rooms and graders.
3. (Checkpoint 2) Midpoint project report (2–3 pages), requiring literature review, training data (if applicable), evaluation data and metrics, baselines, and preliminary results (such as evaluation of baselines). Due 04/09 (Thu). Brief feedback provided by 04/16 (Thu).
4. (Checkpoint 3) Project presentations (04/28 and 04/30) – likely breakout rooms; duration will be finalized based on # of teams
5. (Checkpoint 4) Project report (6–8 pages) – Due 05/08 (Thu)

Overview

The final project is an open-ended research or systems project in Natural Language Processing (NLP) or its intersections with areas like vision, health, HCI, education, security, information retrieval, agentic systems, or social science. Students may (a) design a project from scratch, (b) start from one of the example topics, or (c) modify/extend an existing idea, so long as the project has a clearly defined task and is evaluated with a reproducible experimental protocol.

A successful project should at least match the standard of a mini conference workshop paper: it states a well-scoped problem, motivates why it matters, proposes a method or analysis, and provides evidence through experiments and evaluation. Projects can be model-building, post-training, dataset/benchmark creation, evaluation/analysis tooling, or interactive systems. But they must include standard deliverables which will be mentioned below.

We strongly encourage choosing a scope that can be completed within the course timeframe. Students are free to extend their project after the final deadline and submit to peer review venues. Submit the pdf via gradescope. For the final report, you can submit it along with your public repository or add this email zn.tang.terran@gmail.com to your private repository.

We will later release details on compute resources.

General Project Deliverables

Students are free to design their own projects, simply reproduce an existing paper, or use/start-from/modify the example projects. Still, there are generally deliverables that apply to most project designs. For checkpoints 1 and 2, note that there are required deliverables. **Still, the content can be significantly changed for the final report. Email me if you are considering changing to a completely new topic.** If you need other special accommodations, please email GSI **Zineng Tang**, terran@berkeley.edu.

Checkpoint 1 deliverable:

1) Problem Definition + Scope

Checkpoint 2 deliverable:

2) Dataset / Environment Artifact

1. You can choose
 - New Dataset (collected)
 - Synthetic Dataset (procedurally generated)
 - (Subset of) Existing dataset

3) Baselines (at least 1-2)

1. Zero-shot / prompt-only baseline
2. Heuristic baseline
3. Existing SOTA model or open LLM baseline

4) Literature Review

5) Evaluation Suite + Metrics

1. Primary metric aligned to the task for your baseline (like accuracy)
2. Efficiency (optional)
3. Generalizability (optional)

Final deliverable:

6) Proposed Method / System (1 of 2)

1. Proposed method (could be a model, training paradigm, data generator, etc.)
2. Proposed analysis tool

7) Main Results

8) Ablations (at least 2)

1. Example:
 - a. Remove a key component in your framework

- b. Compare different prompts

9) Error Analysis (optional)

1. Example
 - Insight: “If we fixed X, performance would likely improve”
 - Failure modes or examples

10) Qualitative Demonstrations

1. Example
 - A sample data if using synthetic dataset
 - Example model input or output
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Writeup Requirements

Here is a general writeup structure for a 6-8 page 1 column paper, where references and acknowledgements are not counted. Use this package for either latex or .docx writing: <https://github.com/ICLR/Master-Template/raw/master/iclr2026.zip>. If you plan to use another template or use your own, please notify me to check if they are in good standards. The writeup quality should at least match with standard conference workshop submission.

Below are general guidelines for paper structure but not fully restricted. For example, you can reorganize your approaches into different method sections. If you are considering certain sections to be longer than usual or exceeding the paper page maximum, please email TA **Zineng Tang**, terran@berkeley.edu.

Note: Deliverables for different checkpoints and final reports should be accumulated, meaning that your latest writeup should include all previous checkpoints.

Checkpoint 1 deliverable:

1) Title + Abstract (0.5 to 0.8 page)

- Problem → method/approach → (anticipated) key results → why it matters
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Checkpoint 2 deliverable:

2) Introduction (within 1 page)

3) Related Work (within 1 page)

- Literature review with references included

4) Baseline Setup (within 1 pages, can latter be merged with “6) Method Setup” and “7) Experiment”)

- Implementation details of baseline
- Dataset details or environment setup
- Evaluation setup and baseline results

5) If you would like to be considered for the VESSL AI credits, write a short paragraph on "Application to VESSL AI credits" describing your plans on how to use GPU hours.

Final deliverable:

6) Method Setup (1-3 pages)

- Core idea
- Implementation details (enough to replicate)
- Any theoretical framing

7) Experiment (1-2 pages)

- Hyperparameters, etc.
- Baselines and your method
- Main results table

8) Analysis (within 1 page)

- Ablations
- Error analysis and qualitative examples

9) Limitations and Broader Impact and Ethics if relevant (1-2 paragraphs, optional)

10) Conclusion (1 paragraph)

11) Appendices (optional)

- Additional experiments details like prompts
- Additional qualitative or quantitative results

Reference Paper Topics

Overview: We provide example paper topics for reference covering different NLP and Vision-Language areas. If you intend to select any of the below topics or modify them, feel free to email TA **Zineng Tang**, terran@berkeley.edu for any questions you have.

Fill out this google excel,

https://docs.google.com/spreadsheets/d/1Bhi5m-lyiQ_49CVsYbA_nwNVFJgCfkY_3PCKVKGN0U/edit?usp=sharing with your name/email in the corresponding topic, if you are interested in any of the below topics but couldn't find a team.

Topic 1: Representing Language as World States in Symbolic Text Environments

World models have emerged as a powerful tool for representations in agents. However, language does not explicitly give underlying states. Easily, one could use a tool-calling or web-browsing environment to transform LLM into a flexible agent simulating a world model behavior. However, how do world states work in natural text? For the sentence, "I go to Starbucks and then buy a cappuccino." It can be defined as

1. action_0: go to starbucks, state_0: I'm currently not at a Starbucks
2. action_1: buy a cappuccino, state_1: I'm currently at Starbucks but don't have any coffee

Deliverables:

1. A well-defined action space, observation space, and states.
2. A well-defined task at which the "LLM world model" outperforms "just LLM"

Topic 2: Self-Play LLMs with Evolution Strategies

Post training with RL scales well but requires a lot of data. Recent works have shown that evolution algorithms have promising potential in optimizing data constrained tasks. Intuitively, well-designed multi-LLMs can operate more efficiently on certain tasks. Using the same copies of LLMs can enable more flexible communication. But there are many perspectives to go from